

A Common Ground Consortium case study for Nuclear Siting Tools

R. A. Borrelli · Denia Djokić



University of Idaho
Department of Nuclear Engineering
and Industrial Management



**NUCLEAR ENGINEERING &
RADIOLOGICAL SCIENCES**
**FASTEST PATH
TO ZERO INITIATIVE**
UNIVERSITY OF MICHIGAN

American Nuclear Society Annual Meeting – Denver, Colorado

2026.06.03

Collaboration-Based Siting of Spent Nuclear Fuel

2021 Consolidated Appropriations Act allocated funds for 'expenses necessary for nuclear waste disposal activities ...'

'...including storage activities'

DOE awarded \$26M to thirteen (now twelve) consortia across the nation for building capacity in Collaboration-Based Siting (CBS)

Support more comprehensive community engagement to collaborate with stakeholders to manage SNF effectively

Not actual siting

Process and frameworks

Twelve consortia awarded across the country

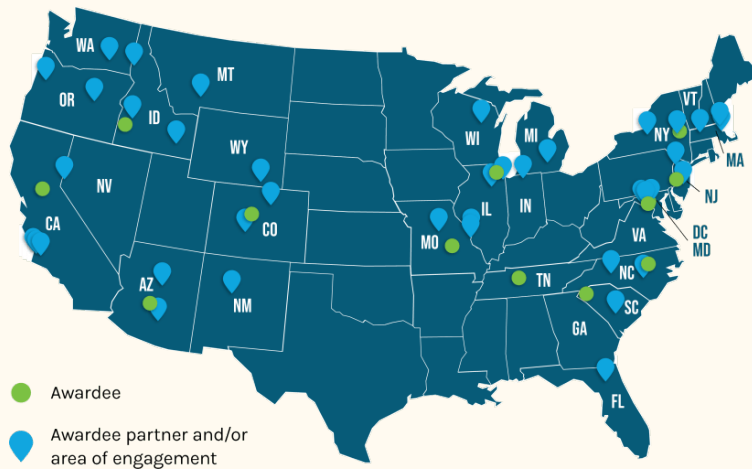


Figure. Nationwide consortia

[1] Department of Energy, 2023. DOE awards \$26 million to support consent-based siting for spent nuclear fuel

The Common Ground Consortium (CGC)

CGC spans several institutions across the intermountain west and Denia at the University of Michigan

With Nation Tribal Energy Association

Ended in March 2026

But we got a NEUP!

Centered on public input

Research focus on historical lessons, types of public decision-making, and state-specific and/or cross-jurisdictional challenges in Collaboration-Based Siting of critical infrastructure

Providing recommendations to DOE

[2] Araújo, K., 2026. From Stalemate to Collaboration: Rethinking Nuclear Waste Siting and Policy. Phoenix, Arizona: Proc., Waste Management Symposium

[3] Koerner, C. et al., 2025. The Willrich report predictions. Washington, D. C.: Proc., IHLRWM

[4] Borrelli, R. A., et al., 2024. The Common Ground Consortium Focus on Public Conversations. Las Vegas, Nevada: Proc., ANS Annual Meeting

Motivation & Goals

Lots of executive orders were issued in 2024 for nuclear energy growth

400 GWe by 2025!

Deploying reactors at military bases

Recycling & reprocessing

Hopefully pyroprocessing for Christina and me

Increasing domestic fuel production

July 4th milestone

All this growth really requires strong back end management policy

More SNF facilities are going to be needed at the least

Communities or localities may want help in terms of technical feasibility, local capacity to host a site

There are some tools out there that could provide site screening information

We critically reviewed and tested some of them within this context

Based on community accessibility and utility just for an SNF Consolidated Storage Facility (CSF) for now

Because we were only funded for that at the time

Subset of a larger study for journal submission

Just four today

Siting Tool for Advanced Nuclear Development (STAND)

STAND examines feasibility for advanced nuclear facility siting

Population, socioeconomics, safety, regulations, etc.

Comparative analysis for different sites

Site Exploration option with economic, proximity, reference, safety data layers

[5] Fastest Path to Zero, 2022. STAND – Quickstart guide. INLRPT-23-74581

STAND example for Bonneville County siting

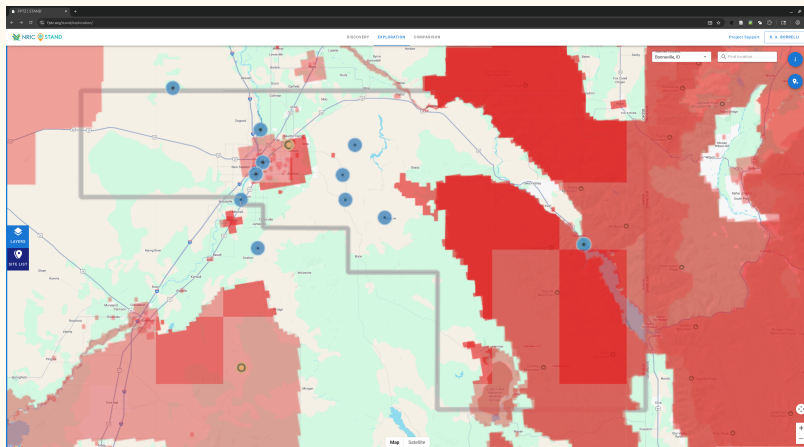


Figure. User generated

Red – protected lands

Blue – Electric
generators

Yellow – Brownfield site

STAND could be useful
for screening

Could use more
information on
regulations

And more on the criteria
themselves

Web-Based Transportation Routing Analysis Geographic Information System (WebTRAGIS)

WebTRAGIS is a GIS web tool developed by ORNL for transportation routing

Databases for highway, rail, waterways

Select origin and destination to generate route

Optimize transportation under distance, preferred routes, time

Layers are county boundaries, critical infrastructure, population, etc.

[6] Peterson, S., 2018. WebTRAGIS: Transportation Routing Analysis Geographic System User's Manual. ORNL/TM-2018/856

WebTRAGIS national population density with railways

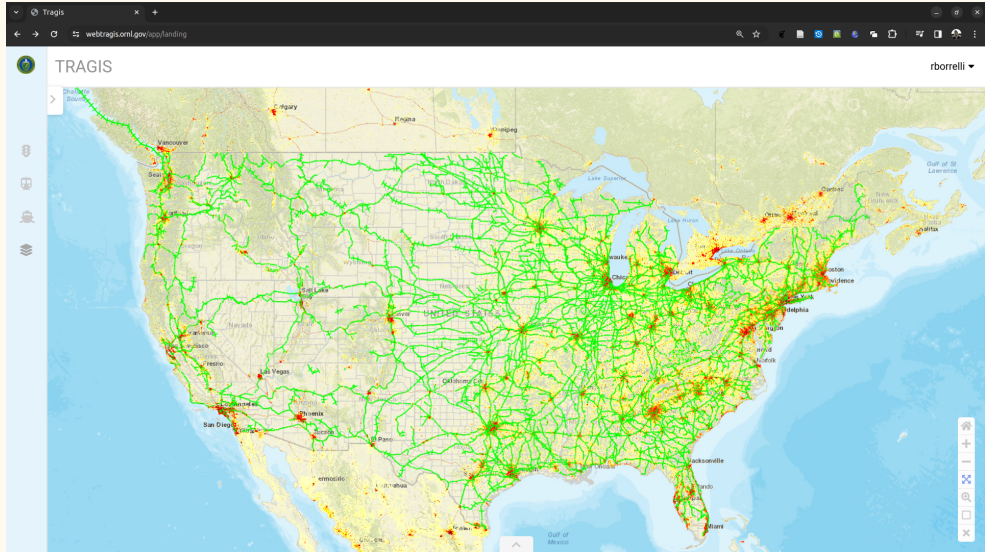
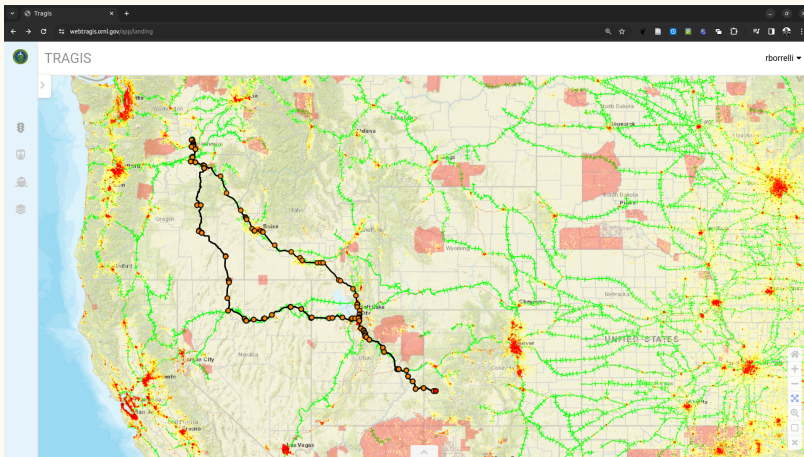


Figure. User generated

WebTRAGIS transport from Richland to Naturita



Southern route includes Idaho
Direct route goes through Idaho
Node based generation
1300 miles & 32 hours

Several data layers
Multiple travel modes
Nodes not shown beforehand; only after destination
Communities might investigate locations of critical infrastructure, population
Communities would not be planning transportation routes though

Figure. User generated

Land-area Identification, Tagging, and Exploration (LITE)

LITE (ORNL) to evaluate areas for storage

Compare up to two locations

Three main areas of functionality –

- Explore – Evaluate areas of interest

- Access – Select assessment considerations

- Compare – Generate assessment reports

[7] Le, T. et al., 2023. Developing digital tools for engagement. Presentation

LITE site screening criteria for Bonneville county

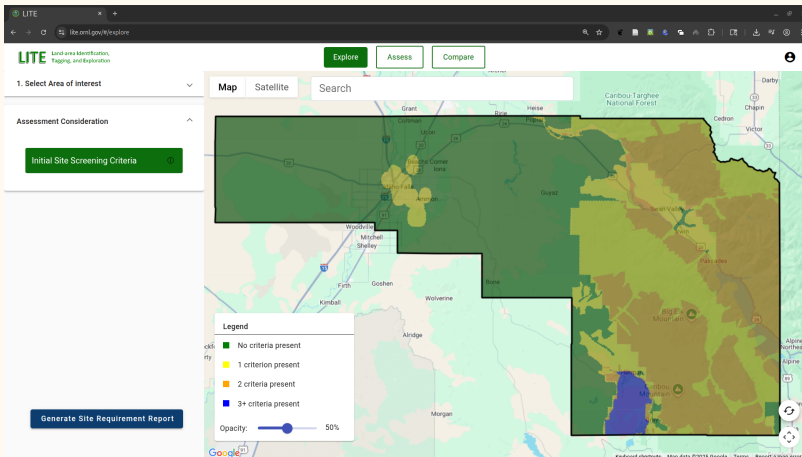


Figure. User generated

Layers include Critical Habitat for Endangered and Threatened Species, Coastal Barriers, Federally Designated Lands, etc.

Energy facilities, floodplains, hazardous facilities

West is wide open

East has been flagged under the criteria

LITE site screening for Bonneville county with all assessment considerations included

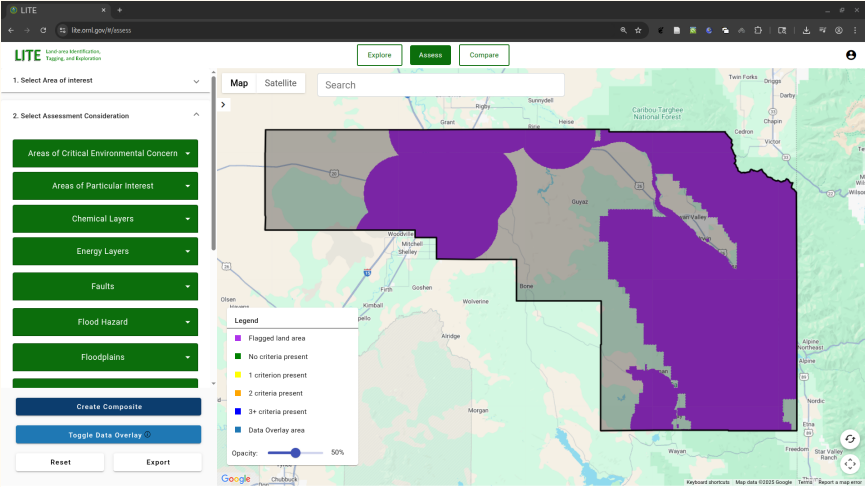


Figure. User generated

LITE comparison for Bonneville county & Southwest Wyoming

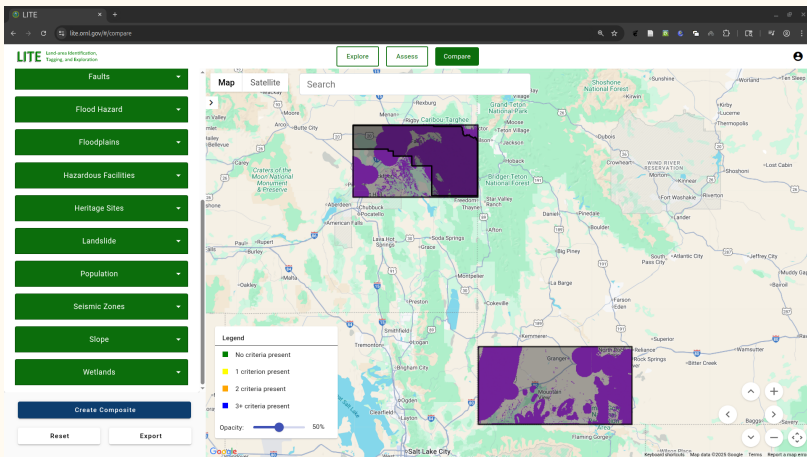


Figure. User generated

Tried to select similar area

Same assessment considerations

Wyoming offers more land due to lower population density

LITE offers limited capabilities

Granularity only down to counties

Comparison feature is useful

Stakeholder Tool for Assessing Radioactive Transportation (START)

START is a GIS decision tool for SNF transport

Designed by DOE

Generates routes by user selected options; e.g., minimizing distance, population, etc.

Select various sites; e.g., infrastructure, police, airports, casinos, etc.

Select origin, destination, transportation method

Multiple transport modes

[8] Abkowitz, M. et al., 2016. Application of a Decision-Support Tool for Evaluating Radioactive Material Transportation Routing Options and Emergency Preparedness. Kobe: Proc., 18th International Symposium on the Packaging and Transportation of Radioactive Materials

START transport from Richland to Pueblo

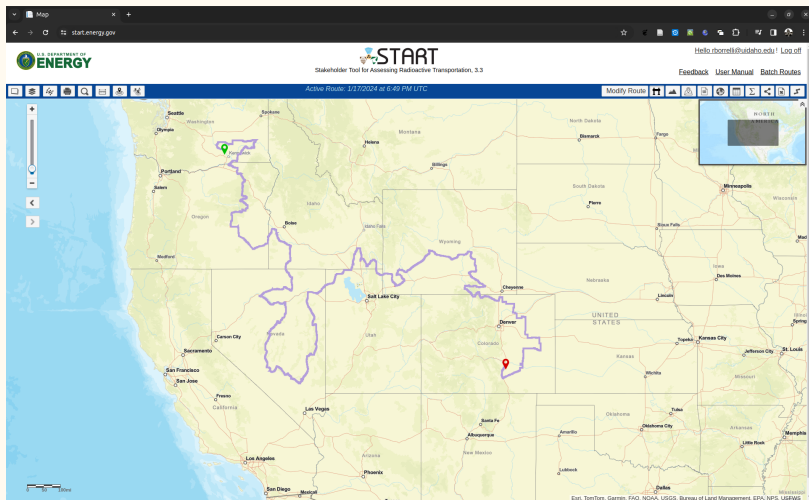


Figure. User generated

Heavy haul truck

Minimize population with
800 m buffer

3500 miles

Windy route due to
buffer

Not much detail

Difficulty in selecting
cities

Could be used to identify
critical sites on a route

Discussion

These tools have drawbacks

Some tools could not be included due to the change in funding mandate

Could not get access to OR-SAGE – Looked promising from literature

A lot of redundancy across all tools studied – Silos

WebTRAGIS & START are basically the same

WebTRAGIS seemed better with options and detail

Same for STAND over LITE

Major drawback is that the sites require registration

Communities may be wary of privacy issues or if the tool owners may be collecting data without consent

Cannot download the images as jpg, png

Use of the tools in the literature was not prevalent

More transparency on data sources should be provided

Lack of open source accessibility

Would be difficult for public at large use without assistance

On the other hand, how could we aid in community decisionmaking?

Site screening and transportation routes do offer use if only just to see critical sites

Could be used for public comments

Community could use tools to develop a go/no-protocol in a siting process

Honestly, not that great overall

Easy to see what fixes are needed

Tools pros & cons I

Table. Summary of tools

Tool	Use	Features	Limits
STAND site	For advanced reactor siting.	Site Exploration methodology could be useful for community planners for screening. Selecting criteria was easy to perform.	Cannot download the map. Criteria lacked sufficient explanations. Registration required.

Tools pros & cons II

WebTRAGIS site	Modeling transportation routing. Uses GIS to identify optimal means of transportation for user selected parameters.	Intuitive GUI allows selection of a lot of layers including population density and Tribal lands. Can combine transport routes manually.	Cannot download the map as a .jpg, etc. National parks, wilderness not included in layers. Cannot just click on the map to plan route; Transportation 'nodes' are not shown on the map. Registration required.
--------------------------------	---	---	--

Tools pros & cons III

LITE site	Evaluate areas of interest for interim storage using a pre-configured assessment considerations.	Presents a map of screening criteria and assessment considerations down to the county level. Areas can also be compared. Descriptions of the criteria and considerations is provided.	Cannot download the map as a .jpg., etc. Lacking granularity. Registration required.
START site	Geospatial decision-support tool developed for evaluating routing options and other aspects of transporting SNF.	Variety of data layers. Printable map options. Can select different transportation modes for one shipping route. Can select for distance, time, population buffer. Can select a wide range of infrastructure to filter.	Hard to select cities, etc., for destinations. Have to zoom in significantly to select a destination. Map renders slowly.

Wrapping up

What is next?

There are a few more tools to check out

Propose a methodology for Collaboration-Based Siting incorporating some of the tools

For effective back-end management, community autonomy can be prioritized by providing the necessary tools for informed, independent decision-making

Finish the journal paper



References

1. Department of Energy, 2023. DOE awards \$26 million to support consent-based siting for spent nuclear fuel.
2. Araújo, K., 2026. From Stalemate to Collaboration: Rethinking Nuclear Waste Siting and Policy. Phoenix, Arizona: Proc., Waste Management Symposium.
3. Koerner, C. et al., 2025. The Willrich report predictions. Washington, D. C.: Proc., IHLRWM.
4. Borrelli, R. A., et al., 2024. The Common Ground Consortium Focus on Public Conversations. Las Vegas, Nevada: Proc., ANS Annual Meeting.
5. Fastest Path to Zero, 2022. STAND – Quickstart guide. INLRPT-23-74581.
6. Peterson, S., 2018. WebTRAGIS: Transportation Routing Analysis Geographic System User's Manual. ORNL/TM-2018/856.
7. Le, T. et al., 2023. Developing digital tools for engagement. Presentation.
8. Abkowitz, M. et al., 2016. Application of a Decision-Support Tool for Evaluating Radioactive Material Transportation Routing Options and Emergency Preparedness. Kobe: Proc., 18th International Symposium on the Packaging and Transportation of Radioactive Materials.

Acronyms I

ANS American Nuclear Society.

CAA Consolidated Appropriations Act.

CBS Collaboration-Based Siting.

CGC Common Ground Consortium.

CSF Consolidated Storage Facility.

DOE United States Department of Energy.

GIS Geographic Information System.

GUI Graphical User Interface.

GWe Gigawatts-electric.

LITE Land-area Identification, Tagging, and Exploration.

NEUP Nuclear Energy University Program.

Acronyms II

OR-SAGE Oak Ridge Siting Analysis for Power Generation Expansion.

ORNL Oak Ridge National Laboratory.

SNF Spent Nuclear Fuel.

STAND Siting Tool for Advanced Nuclear Development.

START Stakeholder Tool for Assessing Radioactive Transportation.

UM University of Michigan.

WebTRAGIS Web-Based Transportation Routing Analysis Geographic Information System.